Code No: 158DP **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B.Tech IV Year II Semester Examinations, September - 2022
MEASURING INSTRUMENTS
(Common to CE, EEE, ME, CSE, EIE, IT)

Time: 3 Hours **Max.Marks:75**

Answer any five questions
All questions carry equal marks
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JNTUH Used Papers

1.a) Explain about sources of errors different types of errors and precautions to minimize them.
b) The accuracy of five digital voltmeters are checked by using each of them to Measure a standard 1.0000V from a calibration instrument. The voltmeter readings are as follows:
V_1=1.001v, V_2=1.002v, V_3=0.999v, V_4=0.998v and V_5=1.0000v.
Calculate the average measured voltage and the average deviation. [7+8]

2.a) Explain the difference between basic standards and secondary standards.
b) Calculate the maximum percentage error in the sum and difference of two voltage measurements when V_1=100v ± 1% and V_2=80v ± 5%. [7+8]

3.a) Derive gauge factor due to change in dimensions of a strain gauge element when it is subjected to tensile force.
b) Calculate the gauge factor S if a 1.5 mm diameter conductor that is 24 mm long Changes length by 1 mm and diameter by 0.02 mm under a compression force. [8+7]

4.a) Explain how LVDT is used to measure linear displacement.
b) Show that a parallel plate capacitor serves as the most suitable transducer for measurement of linear and angular displacements. [7+8]

5.a) Discuss how length is measured using optical method.
b) Explain how roughness is measured accurately. [7+8]

6.a) Explain the velocity measurement method and discuss about the possible errors in measurement.
b) Using multiplexing, explain multi-channel data acquisition system. [8+7]

7.a) Illustrate the principle of force summing devices with an example.
b) What are the main differences in measuring low pressure and high pressure? [7+8]

8.a) Show that a parallel plate capacitor serves as the most suitable transducer for measurement of linear and angular displacements.
b) A transducer that measures force has nominal resting resistance of 300 Ω and is excited by 7.5V. When a 980 dyne force is applied, all four equal resistance bridge elements change resistance by 5.2 Ω . Find the output voltage E_o. [7+8]

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